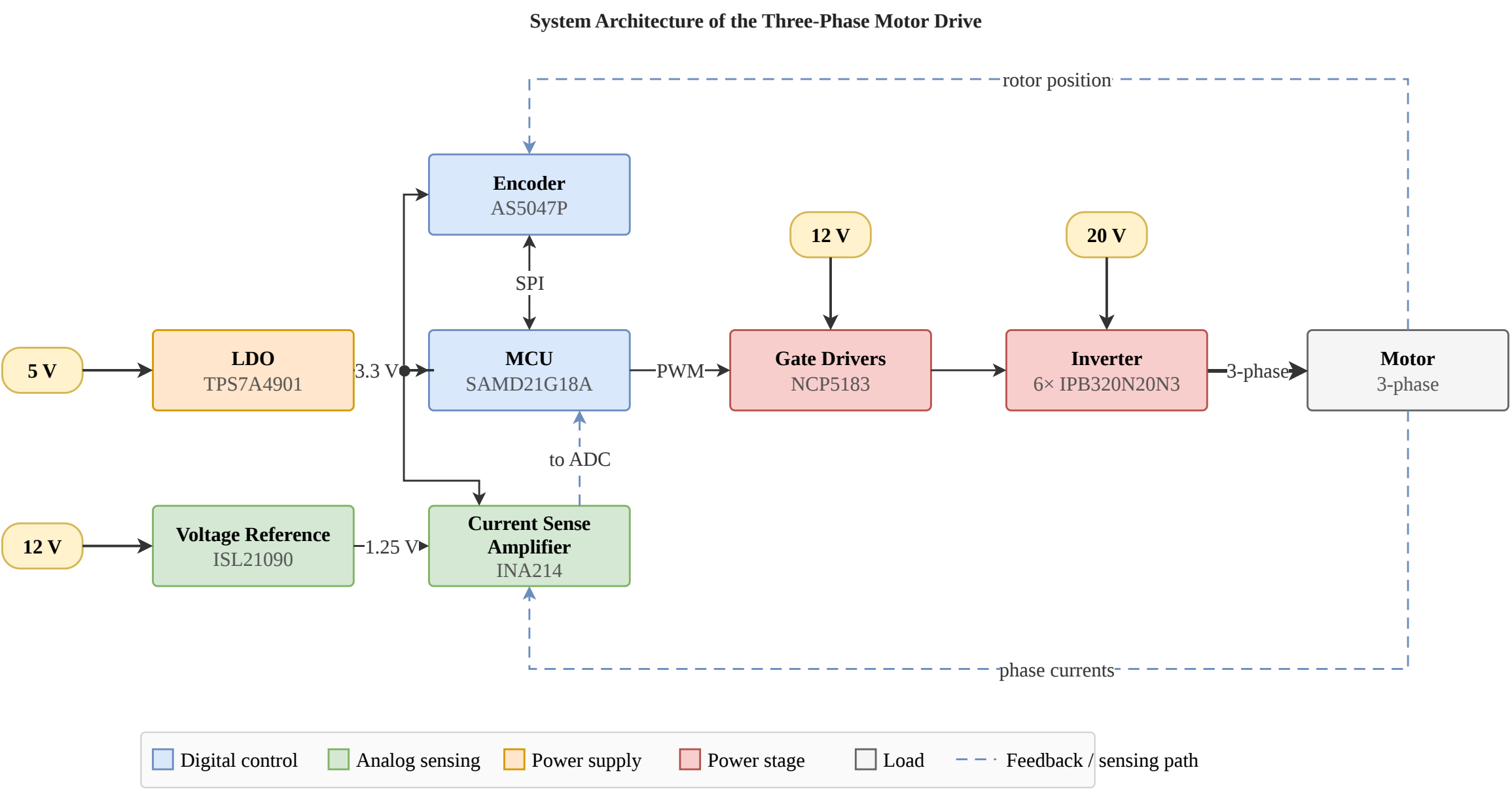
System Architecture of the Three-Phase Motor Drive
rotor position
Encoder
AS5047P
SPI
12 V
20 V
5 V
LDO
TPS7A4901
3.3 V
MCU
SAMD21G18A
PWM
Gate Drivers
NCP5183
Inverter
6× IPB320N20N3
3-phase
Motor
3-phase
to ADC
12 V
Voltage Reference
ISL21090
1.25 V
Current Sense Amplifier
INA214
phase currents
Digital control
Analog sensing
Power supply
Power stage
Load
Feedback / sensing path